cent of its initial length without any change in its power supply properties. The team demonstrated the stretchiness of the battery by embedding it into an elastic armband.

Batteries have always posed a problem for the smart fabrics industry, flexibility being one issue and battery life being another. In order to overcome this, the industry is looking at solutions that enable smart fabrics to harness energy from the environment.

In a recent report in the journal *ACS Nano*, scientists Wenjie Mai, Xing Fan and team have described fibres that can capture and release solar energy. The fibres are suitable for weaving into fabrics that can be tailored normally. The team has devised two types of fibres. One comprises titanium or manganese-coated polymer along with zinc oxide, a dye and an electrolyte. These fibres are interlaced with copper-coated polymer wires to create the solar cell section of the textile.

A second type of fibre helps store power. It is made of titanium, titanium nitride, a thin carbon shell to prevent oxidation and an electrolyte. These fibres can be woven with cotton yarn to make smart garments. The fibres can apparently power small electronics like tablets and phones as well.

Cool fabrics, literally. Can you imagine smart clothes that adjust their temperature automatically according to the wearer's needs, say, to keep a patient comfortable or to cool an athlete's body during intensive workouts? Well, VTT Technical Research Centre of Finland Ltd appears to have a solution, which addresses the thermal, moisture and flow-technical behaviour of smart clothing.

The technology uses VTT's Human Thermal Model calculation tool to calculate a person's individual thermal sensation from collected data and prevailing conditions. The technology can calculate whether and how much the wearer needs to be cooled or warmed. As an example, a smart blanket made with this technology can identify the person's needs, measure the ambient temperature and adjust the blanket's temperature to keep the person comfortable. According to a press release, Taiwan Textile Research Institute has already tested VTT's methods in designing clothing for long-distance runners in different temperatures.

Applications with attitude

A pair of smart shorts or socks is more likely to stay on a hurdle jumper than a smartwatch. A soldier will feel more secure if his or her shirt had sensors and telecommunication equipment than gear in the backpack. An unobtrusive, smart bedsheet is definitely going to be more effective in monitoring the parameters of an aged, senile and bed-ridden patient than a wristband, which he or she will want to remove and perhaps even throw away.

Indeed, there is no need to explain the necessity of making fabrics smarter. Products using smart fabrics, for various purposes, are already available today, and it is interesting to note that elsewhere in the world people included some these products in their Thanksgiving and Christmas shopping lists.

Sports and fitness. Sensoria Fitness has a range of motion- and activity-tracking smart clothing for sports and fitness. From t-shirts to socks and smartbras, they have a wide range of smart clothing. Their socks are infused with textile sensors that can detect foot pressure. Conductive fibres in the socks relay this data to a Bluetooth-powered anklet that communicates with a mobile app, thereby adding a touch of artificial intelligence.

Although initially focused on fitness, Sensoria's solutions have also found a market in healthcare. In association with Orthotic Holdings, Sensoria has developed a solution called Smart MBB, which uses sensors in the plantar region of the foot to detect and prevent falls in elderly patients. Medical journals acclaim